



COMP3055

Machine Learning

Explain the Solution to Lab 2

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2025 Autumn

All imports in this lab

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.datasets import fetch_openml
```

```
from sklearn.neighbors import KNeighborsClassifier
```

Handwritten digit recognition

```
data = np.load('1k.npz', allow_pickle=True)
X_small = data['X_small']
y_small = data['y_small']
index = np.nonzero(y_small == '2')[0]
te_feats = [X_small[i] for i in index]

knn = KNeighborsClassifier(n_neighbors=2)

# Fit the model using X_small as training data and y_small as target values
knn.fit(X_small, y_small)

# Predict the class labels for te_feats (i.e. first 99 images of digit 2)
prediction = knn.predict(te_feats)

print(prediction)
```

Any Questions?

